

Food Delivery Management System

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Course: Data Structures & Algorithms

Department: UBIT

Summary of Project:

Food delivery services receive many customer orders throughout the day. Managing these orders efficiently is important to ensure that customers receive their food in the correct order while allowing staff to quickly locate a specific order when needed. This project simulates a simple food delivery management system using two different data structures.

The system allows users to:

- Place a new order
- Process the next pending order
- Search an order using its Order ID
- Display all pending orders

This project demonstrates how multiple data structures can be combined to solve a real-world problem efficiently.

Data Structures Used:

1. **Queue:** A queue is used to manage customer orders in the order they receive.

- Place Order (Enqueue)
- Cook Next Order (Dequeue)
- Display Pending Orders

2. **Hash Table:** A Hash Table is used to search customer orders quickly by their Order ID without checking every order one by one.

- Insert Order
- Search Order

Tools Used:

Programming Language: C++

IDE/Compiler: Visual Studio Code

Concepts Used:

- Structures
- Dynamic Memory Allocation
- Pointers
- Linked Lists
- Queues
- Hash Tables
- Linear Probing
- Functions
- Arrays

Screenshot:

```
PS D:\C++\_Project\output> cd 'd:\C++\_Project\output'
PS D:\C++\_Project\output> & .\'FoodDeliverySystem.exe'
Order #101 placed successfully.
Order #102 placed successfully.
Order #103 placed successfully.
Pending Orders:
OrderID: 101 | Customer: Someone | Food: Burger
OrderID: 102 | Customer: Another one | Food: Pizza
OrderID: 103 | Customer: Third one | Food: Fries
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Order Found: Customer: Another one | Food: Pizza
Cooking Order #101 for Someone
Pending Orders:
OrderID: 102 | Customer: Another one | Food: Pizza
OrderID: 103 | Customer: Third one | Food: Fries
-----
PS D:\C++\_Project\output> █
```